

CES 2025 Benchmark: First Results

Six fine-tuned models · 150 subgroup scores each · what to keep

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In brief. Fine-tuning works: our best model is about **twice as close** to real CES 2025 subgroups as the same model untrained. The best recipe is **fixed context questions with 50k training examples**. More data (100k) did not help. We are still about 12 points above the human noise floor, and immigration is the weakest item by far.

1. What we tested

Six Llama 3.3 70B models, fine-tuned on 8 Canadian surveys (CES 2019 and 2021, Democracy Checkup 2019–2024). None of them saw any CES 2025 data. They differ on two points:

- **Context format.** At training, each example shows 9 other answers from the same respondent. *Fixed*: the same 9 themes every time (ideology, party, trust...). *Semantic*: the 9 questions closest in meaning to the target.
- **Training size.** 20k, 50k or 100k examples. Each larger set contains the smaller one.

Every model answered the same frozen set: 15,000 synthetic respondents from the Democracy Checkup 2024 (5 regions × 3 age groups), 10 CES 2025 questions, 150,000 calls. We compare the answer distribution of each of the 150 subgroup × question pairs with the real CES 2025 one.

How to read the score. Total variation (TV) is the share of answers that would have to change to match the real distribution. A TV of 0.18 means about **18 answers out of 100** are in the wrong place. Lower is better.

2. Results

Model	Mean TV (150 pairs)	Answers out of place, per 100
Human floor (two halves of CES 2025)	0.062	6
Fixed, 50k	0.178	18
Semantic, 50k	0.188	19
Fixed, 100k	0.191	19
Semantic, 100k	0.195	20
Fixed, 20k	0.200	20
Semantic, 20k	0.214	21
Untrained Llama 3.3 70B	0.325	33
Same answer for everyone (most common one)	0.540	54

Three takeaways.

1. **Training matters most.** Going from untrained to the best model removes 15 out-of-place answers per 100 (0.325 → 0.178). That is more than half the distance to the human floor.
2. **Fixed context beats semantic context** at every size: by 1.4 points at 20k, 1.1 at 50k and 0.4 at 100k. Fixed context is also simpler: the same questions for every item, and no risk of showing a near-copy of the target.

3. **50k is the sweet spot.** 20k → 50k helps (about -2.5 points). 50k → 100k hurts slightly for both formats (+1.4 points fixed, +0.7 semantic).

Each of these comparisons is statistically clear on our test set (paired bootstrap, 95% intervals exclude zero). Invalid replies are rare (under 1% for every model).

Where the best model struggles

Question	Fixed, 50k	Untrained	Human floor
Spending: affordable housing	0.10	0.15	0.05
Satisfaction with democracy	0.12	0.28	0.05
Medical assistance in dying	0.14	0.34	0.09
Jobs before environment	0.15	0.40	0.10
Spending: defence	0.17	0.23	0.04
Pipelines	0.17	0.43	0.10
Spending: national childcare	0.18	0.40	0.06
Personal finances, past year	0.21	0.25	0.04
Spending: reconciliation	0.24	0.45	0.05
Immigration levels	0.30	0.32	0.04

Immigration barely improves with training. It is also the topic most absent from training: Azure’s content check rejected most immigration items. Regions and age groups behave alike (0.17–0.18), except the Atlantic provinces (0.20), which also have the smallest real samples.

3. Caveats

- **One training run per recipe.** The intervals cover the test data, not training luck. We do not yet know how much a second run of the same recipe would move the score. Gaps of about one point (50k vs 100k, fixed vs semantic) need a repeat before we call them real.
- **One base model.** All six use Llama 3.3 70B. The ranking may differ for smaller models.

4. Next steps

1. **Keep “Fixed, 50k”** as our reference model.
2. **Add Justin’s model** (Qwen3-4B). It cannot run in our Azure setup (no GPU access there), so it would run on his Mac with the same frozen file and the same scoring script.
3. **Repeat Fixed 50k and Fixed 100k** with a new seed to check that the size effect holds.
4. **Immigration:** find a way to get immigration questions past the training filter, or report it as a known blind spot.